

Machine Learning for Natural Language Processing

Transformer Representation Models



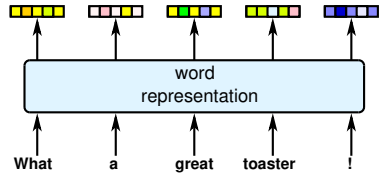
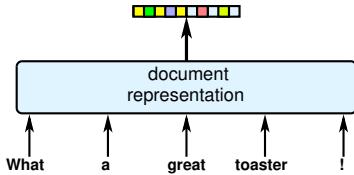
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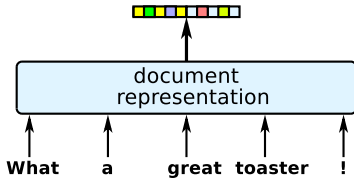
Richard Johansson

`richajo@chalmers.se`

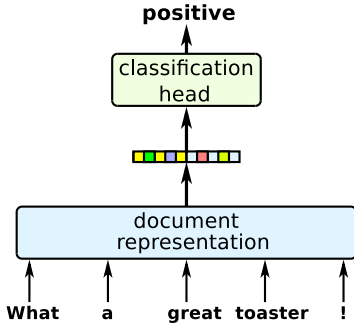
Document and word representations



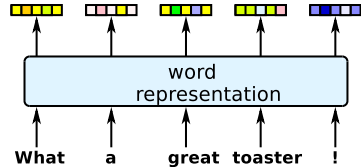
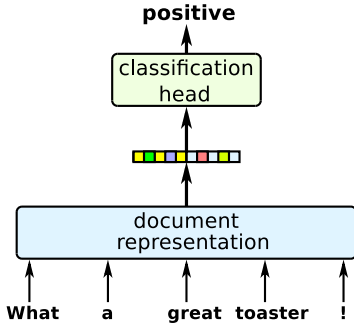
Transfer learning: Pre-training and fine-tuning



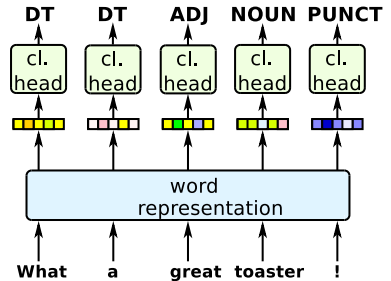
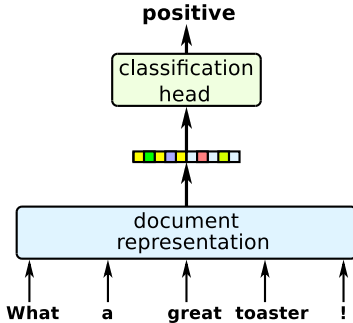
Transfer learning: Pre-training and fine-tuning



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Transfer learning: Pre-training and fine-tuning



Other uses of pre-trained representation models

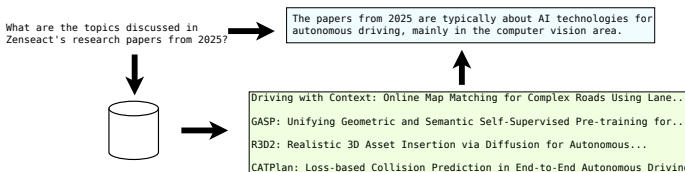
Document vectors can be used to compute the **similarity** between two documents

Very common use case: **retrieval** systems, included **retrieval-augmented generation** models

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Personal / Default project

Dashboard

Q Search

Fine-tuning >

Graders

Specialized models

Image generation

Video generation

Text to speech

Speech to text

Deep research

Embeddings

Moderation

Coding agents

Codex cloud

Agent internet access

Codex CLI

Cookbook

Forum

Vector embeddings

Learn how to turn text into numbers, unlocking use cases like search.

Copy page

New embedding models

text-embedding-3-small and text-embedding-3-large, our newest and most performant embedding models, are now available. They feature lower costs, higher multilingual performance, and new parameters to control the overall size.

What are embeddings?

OpenAI's text embeddings measure the relatedness of text strings. Embeddings are commonly used for:

- Search (where results are ranked by relevance to a query string)
- Clustering (where text strings are grouped by similarity)
- Recommendations (where items with related text strings are recommended)
- Anomaly detection (where outliers with little relatedness are identified)
- Diversity measurement (where similarity distributions are analyzed)
- Classification (where text strings are classified by their most similar label)

Overview

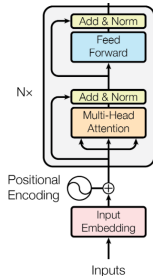
Models

Use cases

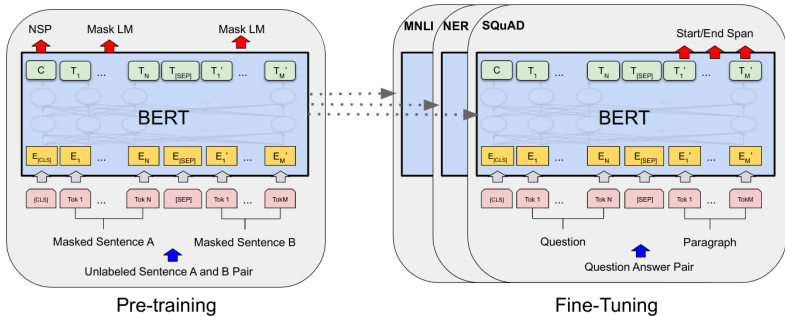
FAQ

BERT (Devlin et al., 2019)

- **BERT** (**B**idirectional **E**ncoder **R**epresentations from **T**ransformers) is an architecture for transfer learning
- Pushed the state of the art for several tasks
- Uses the **encoder** part of the Transformer (originally 12 or 24 layers)
- Key technical point: attention is **bidirectional**



Transfer learning with BERT



Pre-training tasks for BERT (1)

He bought two cans of fish soup .

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He bought two cans of fish soup .

He [MASK] two cans of fish [MASK] .

Pre-training tasks for BERT (1)

He bought two cans of fish soup .

He [MASK] two cans of fish [MASK] .



bought



soup

Pre-training tasks for BERT (2)

The man went to the store. He bought a bottle of milk.



Adjacent

Pre-training tasks for BERT (2)

The man went to the store. He bought a bottle of milk.



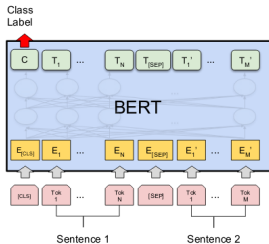
Adjacent

The man went to the store. Penguins are flightless birds.

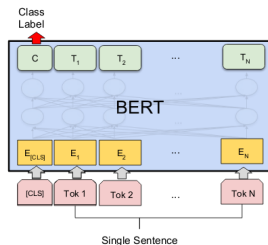


NonAdjacent

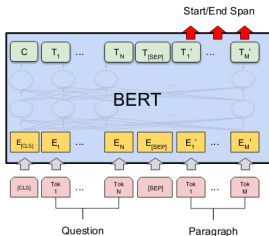
How to use BERT in different types of tasks



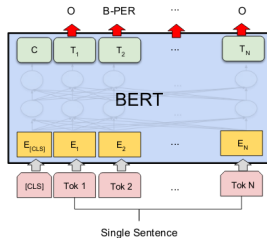
(a) Sentence Pair Classification Tasks:
MNLI, QQP, QNLI, STS-B, MRPC,
RTE, SWAG



(b) Single Sentence Classification Tasks:
SST-2, CoLA



(c) Question Answering Tasks:
SQuAD v1.1



(d) Single Sentence Tagging Tasks:
CoNLL-2003 NER

Other Transformer representation models

- **Improved BERT variants:** RoBERTa...
- **Smaller BERT variants:** DistilBERT...
- **Domain-specific BERT variants:** BioBERT, SciBERT...
- **BERT for improved document representation:** SBERT...
- **Multimodal representation models:** CLIP ...

The MTEB benchmark (Muennighoff et al., 2023)

MTEB: Massive Text Embedding Benchmark

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Abstract

Text embeddings are commonly evaluated on a small set of datasets from a single task not covering their possible applications to other tasks. It is unclear whether state-of-the-art embeddings on semantic textual similarity (STS) can be equally well applied to other tasks like clustering or reranking. This makes progress in the field difficult to track, as various models are constantly being proposed without proper evaluation. To solve this problem, we introduce the Massive Text Embedding Benchmark (MTEB). MTEB spans 8 embedding tasks covering a total of 58 datasets and 112 languages. Through the benchmarking of 33 models on MTEB, we establish the most comprehensive benchmark of text embeddings to date. We find that no particular text embedding method dominates across all tasks. This suggests that the field has yet to converge on a universal text embedding method and scale it up sufficiently to provide state-of-the-art results on all embedding tasks. MTEB comes with open-source code and a public leaderboard at <https://github.com/embeddings-benchmark/mteb>.

Gurevych, 2019) solely evaluate on STS and classification tasks, leaving open questions about the transferability of the embedding models to search or clustering tasks. STS is known to poorly correlate with other real-world use cases (Neelakantan et al., 2022; Wang et al., 2021). Further, evaluating embedding methods on many tasks requires implementing multiple evaluation pipelines. Implementation details like pre-processing or hyperparameters may influence the results making it unclear whether performance improvements simply come from a favorable evaluation pipeline. This leads to the “blind” application of these models to new use cases in industry or requires incremental work to reevaluate them on different tasks.

The Massive Text Embedding Benchmark (MTEB) aims to provide clarity on how models perform on a variety of embedding tasks and thus serves as the gateway to finding universal text embeddings applicable to a variety of tasks. MTEB consists of 58 datasets covering 112 languages from 8 embedding tasks: Bibtex mining, classification, clustering, pair classification, reranking,

Multiple tasks including **similarity, separability, clustering, ...**

The MMTEB benchmark (Enevoldsen et al., 2025)

The screenshot shows the 'Embedding Leaderboard' section of the MMTEB benchmark. On the left, a sidebar under 'General Purpose' includes buttons for 'Multilingual' (selected), 'English', 'Human Benchmark', and a list of categories: 'Image', 'Domain-Specific', 'Language-specific', and 'Miscellaneous'. Below this is a 'Retrieval' section. The main content area is titled 'MTEB(Multilingual, v2)' and contains a description: 'This leaderboard compares 100+ text and image embedding models across 1000+ languages. We refer to the publication of each selectable benchmark for details on metrics, languages, tasks, and task types. Anyone is welcome to add a model, add benchmarks, help us improve zero-shot annotations or propose other changes to the leaderboard.' It also lists statistics: 'Number of languages: 1038', 'Number of tasks: 131', 'Number of task types: 9', and 'Number of domains: 28'. On the right, there are links to 'Cite and share this benchmark', 'Customize this Benchmark', and 'Advanced Model Filters'. At the bottom, a navigation bar includes 'Summary' (underlined), 'Performance per Model Size', 'Performance per Task Type', 'Performance per task', and 'Task information'.

Originally 8 tasks, only English documents

New version: 131 tasks, 1038 languages

Takeaways

A **representation model** maps a document or a word to a numerical representation

Prototypical representative: **BERT**

A relevant **pre-training task** is required

Originally, main focus was **transfer learning**: pre-training and specialization

More recently, the main use case is in **document retrieval** and other similarity-based tasks

References

- J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova. 2019. [BERT: Pre-training of deep bidirectional transformers for language understanding](#). In *NAACL*.
- K. Enevoldsen, I. Chung, I. Kerboua, and M. Kardos et al. 2025. [MMTEB: Massive multilingual text embedding benchmark](#). In *ICLR*.
- N. Muennighoff, N. Tazi, L. Magne, and N. Reimers. 2023. [MTEB: Massive text embedding benchmark](#). In *EACL*.